

Decent Coaching Classes
IX CBSE
Mind Your Math - IV
Answer Key

Section A – Multiple Choice Questions

(5)

1. (D) $\sqrt{7}$
2. (A) 0 [A non-zero constant polynomial has degree 0]
3. (A) $(a - b)(a^2 + ab + b^2)$
4. (B) bisects
5. (A) $7/9$

Section B – Very Short Answer Questions

(8)

1. Classify as rational or irrational:

- (i) $\sqrt{16} = 4$, which is a whole number that can be written as $4/1 \Rightarrow$ rational.
(ii) $\sqrt{11}$ — 11 is not a perfect square, so $\sqrt{11}$ is non-terminating, non-repeating $\Rightarrow$ irrational.

2. Let $x = 0.666\ldots$

$$10x = 6.666\ldots$$

$$\text{Subtracting: } 10x - x = 6.666\ldots - 0.666\ldots \Rightarrow 9x = 6$$

$$\therefore x = 6/9 = 2/3$$

3. $p(x) = 3x - 5$

$$\begin{aligned} p(-2) &= 3(-2) - 5 = -6 - 5 \\ &= -11 \end{aligned}$$

4. $998^2 = (1000 - 2)^2$

$$\begin{aligned} \text{Using } (a - b)^2 &= a^2 - 2ab + b^2, \text{ with } a = 1000, b = 2: \\ &= 1000^2 - 2(1000)(2) + 2^2 = 1,000,000 - 4000 + 4 \\ &= 996,004 \end{aligned}$$

OR

$$\text{Factorise: } x^2 + 9x + 18$$

$$\text{Factors of 18 with sum 9: 6 and 3}$$

$$= (x + 6)(x + 3)$$

Section C – Short Answer Questions

(9)

1. Proof that $\sqrt{3}$ is irrational (by contradiction):

Let us assume, to the contrary, that $\sqrt{3}$ is rational.

Then $\sqrt{3} = a/b$, where a and b are integers with no common factor other than 1 ($b \neq 0$).

So, $a = \sqrt{3} b$. Squaring both sides: $a^2 = 3b^2 \dots (1)$

$\therefore 3$ divides a^2 , so 3 divides a . Let $a = 3c$ for some integer c .

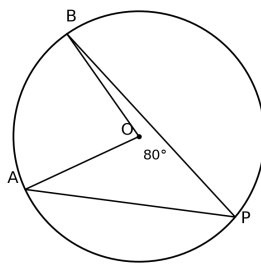
$$\text{Substituting in (1): } (3c)^2 = 3b^2 \Rightarrow 9c^2 = 3b^2 \Rightarrow b^2 = 3c^2$$

$\therefore 3$ divides b^2 , so 3 divides b .

Thus, both a and b are divisible by 3, which contradicts the assumption that a and b have no common factor other than 1.

$\therefore$ Our assumption is wrong, and $\sqrt{3}$ is irrational.

2. Diagram-based question



The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining (major) arc of the circle.

$$\therefore \angle AOB = 2 \times \angle APB$$

$$80^\circ = 2 \times \angle APB$$

$$\therefore \angle APB = 40^\circ$$

3. Expand: $(3a + 2b - 1)^2$

Using $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2xz$, with $x = 3a$, $y = 2b$, $z = -1$:

$$= (3a)^2 + (2b)^2 + (-1)^2 + 2(3a)(2b) + 2(2b)(-1) + 2(3a)(-1)$$

$$= 9a^2 + 4b^2 + 1 + 12ab - 4b - 6a$$

OR

$$\text{Factorise: } 8p^3 + 27q^3 = (2p)^3 + (3q)^3$$

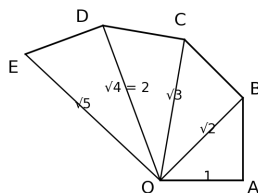
Using $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$, with $a = 2p$, $b = 3q$:

$$= (2p + 3q)(4p^2 - 6pq + 9q^2)$$

Section D – Long Answer Questions

(8)

1. Construction — Square-root spiral



Steps:

(i) Draw a line segment $OA = 1$ unit.

(ii) At A, draw $AB \perp OA$ with $AB = 1$ unit. Join OB.

By the Baudhāyana–Pythagoras theorem: $OB^2 = OA^2 + AB^2 = 1^2 + 1^2 = 2 \Rightarrow OB = \sqrt{2}$.

(iii) At B, draw $BC \perp OB$ with $BC = 1$ unit. Join OC.

$$OC^2 = OB^2 + BC^2 = 2 + 1 = 3 \Rightarrow OC = \sqrt{3}.$$

(iv) At C, draw $CD \perp OC$ with $CD = 1$ unit. Join OD.

$$OD^2 = OC^2 + CD^2 = 3 + 1 = 4 \Rightarrow OD = \sqrt{4} = 2.$$

(v) At D, draw $DE \perp OD$ with $DE = 1$ unit. Join OE.

$$OE^2 = OD^2 + DE^2 = 4 + 1 = 5 \Rightarrow OE = \sqrt{5}.$$

(vi) With O as centre and radius OE, draw an arc to cut a fixed number line at point P. Then $OP = \sqrt{5}$, so P represents $\sqrt{5}$ on the number line.

2. $h(w) = 3w + 12$

(i) $h(w) = 3w + 12$

(ii) $h(6) = 3(6) + 12 = 18 + 12 = 30 \text{ cm}$

(iii) Table of values:

$w = 0 \rightarrow h = 12$; $w = 2 \rightarrow h = 18$; $w = 4 \rightarrow h = 24$; $w = 6 \rightarrow h = 30$; $w = 8 \rightarrow h = 36$

Plotting these points (0,12), (2,18), (4,24), (6,30), (8,36) and joining them gives a straight line graph.

From the graph, $h = 30 \text{ cm}$ at $w = 6$.

$\therefore$ The plant will be 30 cm tall after 6 weeks.

— *End of Answer Key* —